Lecture 4; Dimensions and Diffusion

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Sugars/hour/coli

How many glucose molecules does an *E. coli* cell consume in an hour.

- $10^3 - 10^5$ glucose/hr
- $10^5 - 10^7$ glucose/hr
- $10^7 - 10^9$ glucose/hr
- $10^9 - 10^{11}$ glucose/hr
- what conditions are you talking about?

Let's start with what we know. A bacterial cell is made (to a large extent) of protein.

$$f \times 10^6 \text{ proteins/cell} \times f \times 10^2 \text{ AA/protein} \times f \text{ C/AA} \approx f \times 10^9 \text{ C atoms/cell}$$

Since protein makes up $\approx 50\%$ of cell dry mass, we end up with $\sim 10^{10}$ C/cell.

Rule of Thumb: C atoms in bacteria

$\approx 10^{10}$ carbon atoms in a bacterial cell.

E. coli cells can double in 30-60 min on glucose, i.e., 1-2 doublings per hour. Let's assume 1 hr.

$$\frac{10^{10}\text{C atoms}}{1 \text{ hr}} \times \frac{f \text{ C taken up}}{1\text{C to cell}} \times \frac{1\text{glucose}}{6 \text{ C}} \approx 10^{10}\text{glucose/hr} \approx f \times 10^6\text{glucose/s}$$

The bacterium has to encounter at least 10^6 sugars in every second in order to take them up and eat them. **How does that happen?**

2 Null models

What's our most naive answer to the question of how bacteria encounter carbon? This is our null model, the one we must disprove to argue for a more complex explanation.

What is it here? It's **randomness**. Random movement of molecules at finite temperature, or **diffusion**. We might imagine something more complex, like swimming and chemotaxis, but there are bugs that don't do those things and they may not be needed to explain what's doing on.

3 Diffusion / Brownian Motion

In 1827 Robert Brown reported his observation of pollen grains under a microscope. He subtitled his manuscript "on the existence of active molecules."

> The grains of pollen ... were filled with particles or granules of unusually large size, varying from nearly $\frac{1}{4000}$th to about $\frac{1}{3000}$th of an inch in length ... I observed many of them very evidently in motion.[1] — Robert Brown, 1827

We now know **the molecules don't need to be active** themselves. Even inert molecules are "kicked around" by thermal energy. Random kicking results in random molecular motion **with no directional preference**. This is *diffusion* or Brownian motion. Einstein (1905) provided a theoretical explanation of random

[1] So how big are pollen grains? $\frac{1 \text{ inch}}{f \times 10^3} \approx \frac{f \text{ cm}}{f \times 10^3} = 10^{-5}$ m or about 10 microns in diameter

Outline for whiteboard

- mention HW
- glucose/hr
- diffusion quiz
- null models
- brown quote / perrin
- fick's #1
- units & signs
- dimensions & units
- diff. time - DIY
- dim. analysis
- emergence
- coin tossing
- D scaling / measure
- glucose t_{diff}
- protein t_{diff}
- perfect abs
- teaser: perfect enz

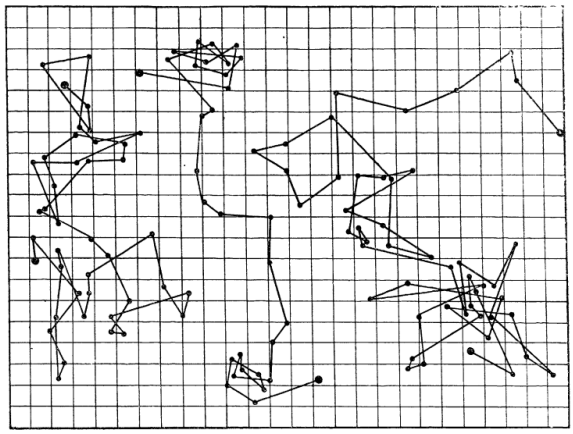

Figure 1: Tracings of the motions of three colloidal particles of radius 0.53 microns as seen under the microscope (Perrin 1909). Each point representing that particle's successive position every 30 seconds; the points are then joined by straight line segments. Caption text reproduced from Wikipedia on Brownian motion

motion via the kinetic energy of atoms & molecules by. We will cover the predictions of this theory today. They were verified experimentally by Jean Perrin in 1909 (see Figure 1).

Here are some nice videos to get you in the mood:

1. Replicating Brown's 1827 observation
2. Similar but with more commentary

4 Diffusion quiz: how fast?

How long does it take a molecule of glucose to diffuse across an *E. coli* cell?

1. less than a millisecond
2. 1-10 ms ✓
3. 10-100 ms
4. 100 ms - 1 s
5. more than 1 s

We'll talk about diffusion to learn how to arrive at this answer.

5 Fick's first law

We are thinking about molecules (of glucose) crossing a boundary (the cell membrane), as in Figure 2. How fast can this happen?

We have an intuition that the flux J_x across a plane is related to the concentration difference on opposing sides.

$$J_x \propto -\frac{\Delta c}{\Delta x}$$

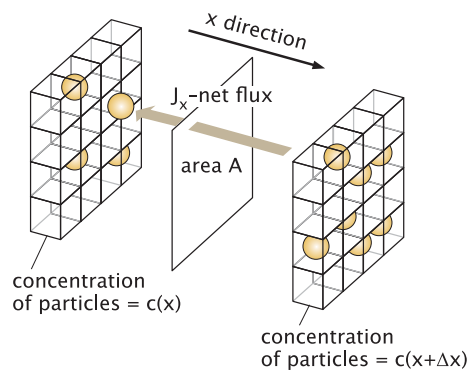

Figure 2: Random motion of a molecule between adjacent planes.

In the limit of small Δx we'll write

$$J_x = -D\frac{dc}{dx}$$

where D is a constant of proportionality called the diffusion constant.

5.1 Whence the minus sign?

Why is there a minus sign? Because stuff flows towards lower concentration. So when concentration is increasing with spatial coordinate x, the molecules flow in the opposite direction (i.e., towards lower x, see Figure 13).

We can infer the units of D by making the LHS and RHS dimensions match. J_x has dimensions of number (#) / time / area (L^2) or $\frac{\#}{L^2 T}$ while $\Delta c/\Delta x$ has dimensions of molecules / volume (L^3) / length (L) or $\frac{\#}{L^4}$. D therefore has dimensions of $\frac{L^2}{T}$, e.g., $\mu\mathrm{m}^2\,\mathrm{s}^{-1}$.

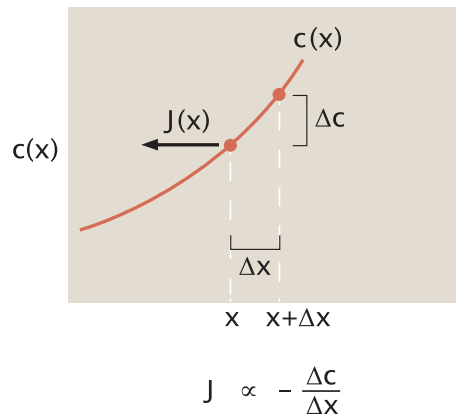

Figure 3: Increasing concentrations means negative flux.

6 Dimensions & Units

In "Streetfighting Mathematics," Sanjoy Mahadevan gives an example from a critique of multinational corporations

> In Nigeria, a relatively economically strong country, the GDP [gross domestic product] is $99 billion. The net worth of Exxon is $119 billion. "When multinationals have a net worth higher than the GDP of the country in which they operate, what kind of power relationship are we talking about?" asks Laura Morosini.
> – Anne Marchand. Impunity for multinationals. September 2002.

What's wrong with this comparison? Is it another example of numbers as "decorations?"

GDP has units of [$/year] and net worth has units of [$]. They are both denominated in $ but they don't have the same **dimensions**. It would make much more sense to compare a quantity with dimensions [money/time] to another with the same dimensions.

Another way of saying this: we could denominate GDP in [euros/day] and net worth in [yen]. We would need to convert these into the same currency to make a comparison, but even then the comparison would be suspect **the dimensions don't match**.

7 Diffusion times by dimensional analysis

Back to our quiz question.

We want to calculate the diffusion time t_{diff}. This is some function of length l and diffusion constant D. Posit a relationship

$$t_{diff} \propto D^{\alpha} \cdot l^{\beta}.$$

Can you use the lines of D and l to derive values for α and β? Are they unique? **Try it for yourselves.**

The exponents α and β are placeholders that let us match the units on the LHS and RHS. Lets write it again with the units.

$$T \propto \left(\frac{L^2}{T}\right)^{\alpha} \cdot L^{\beta}.$$

What values work here? If we set $\alpha = -1$ and $\beta = 2$ we get the right units. It follows that

$$t_{diff} \propto l^2/D \text{ or}$$
$$l_{diff} \propto \sqrt{D \cdot t}.$$

7.1 A more formal approach

We know that the dimensions on the left and right must equal each other. On the left we have $T^1 L^0$ while on the right we have $T^{-\alpha} L^{2\alpha+\beta}$. We can separate this into two equations, one for each dimension

$$T^1 = T^{-\alpha} \implies \alpha = -1 \qquad L^0 = L^{2\alpha+\beta} \implies \beta = -2\alpha = 2$$

Whether we solve for α abd β formally or by trial and error, this approach to *guessing the form of mathematical relationships from the dimensions of the quantities related* is called **dimensional analysis**.

We derived that $t_{diff} \propto D \cdot l^2$. **This relationship is a proportionality.** The dimensions match, but we are missing a dimensionless constant of proportionality. In 3D the constant is 6.

$$t_{diff} = \frac{1}{6}\frac{l^2}{D} \tag{1}$$
$$l_{diff} = \sqrt{6Dt} \tag{2}$$

The only thing that changes between 1D, 2D, and 3D is this prefactor (6 in 3D). The book has a nice vignette with some additional detail.

8 Emergent phenomena

What does $l^2 \propto Dt$ mean? Think about the jiggling particles of Perrin and Brown. An individual particle could move much faster by chance. Collectively, however, they spread out a distance $l \propto \sqrt{(}t)$ **on average**. This relationship is an *emergent property* of considering many particles. To see this clearly, its helpful to think about the random motion of particles more explicitly, as if governed by coinflips.

9 1D Diffusion as Coin Flips

Lets talk about randomness by treating diffusion as coin flips in 1D.

Our walker starts out at $x = 0$. At each discrete time t we flip a coin and she either moves left a units or right a units (a is the step size).

$$x(t) = x(t-1) + \delta x$$

where $\delta x = \pm a$ as in Figure 4.

t	Positions (probability)	$\langle x \rangle$	$\langle x^2 \rangle$
0	$0\ (1)$	0	0
1	$-a\ (\frac{1}{2}),\ +a\ (\frac{1}{2})$	0	a^2
2	$-2a\ (\frac{1}{4}),\ 0\ (\frac{1}{2}),\ +2a\ (\frac{1}{4})$	0	$2a^2$
3	$-3a\ (\frac{1}{8}),\ -a\ (\frac{3}{8}),\ +a\ (\frac{3}{8}),\ +3a\ (\frac{1}{8})$	0	$3a^2$

Table 1: Mean and mean-square displacement for a 1D random walk with step size a. By symmetry $\langle x \rangle = 0$ at all times; calculating $\langle x^2 \rangle_t = x^2 \cdot p(x,t)$ suggests $\langle x^2 \rangle = a^2 t$ as inferred from dimensional analysis.

What is the probability $p(x,t)$ of being at position x at time t? We can think this through with the help of the diagram in Figure 4.

At each step, the mean displacement $\langle x \rangle = 0$ while the **mean squared displacement** $\langle x^2 \rangle \propto t$. The root mean squared displacement $RMSD = \sqrt{\langle x^2 \rangle} \propto \sqrt{t}$ as we inferred by dimensional analysis. Below I describe two ways of proving this relationship for all t.

How can it be that $\langle x \rangle = 0$ while $\langle x^2 \rangle \propto t$? It helps to consider the probability distribution $p(x,t)$ over time. Since the walker is equally likely to move left as right, the distribution is always symmetric about 0 and always has $\langle x \rangle = 0$. Yet, as we can see in Table 1 and Figure 5, the distribution spreads out with time and the walker is increasingly likely to be far from the origin.

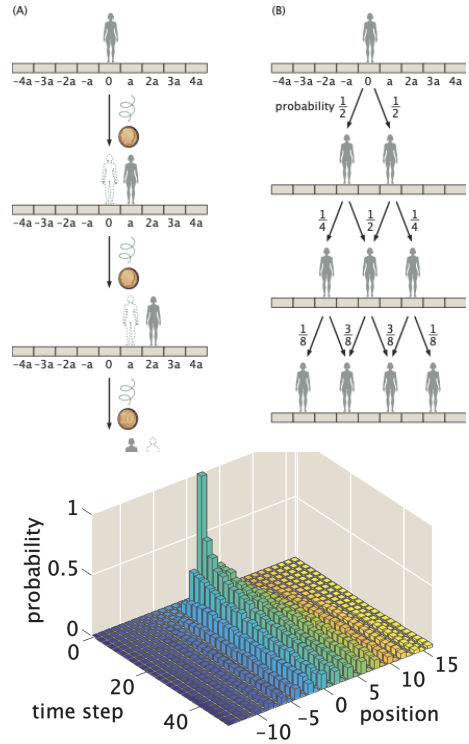

Figure 5: Simulation of 1D diffusion over time via Rob Phillips and Nigel Orme.

9.1 A inductive approach to $l^2 \propto t$ in 1D

We can write the squared displacement at time t as a function of displacement at time $t-1$ and the step size a.

$$x(t)^2 = (x(t-1)+a)^2 = x(t-1)^2 + x(t-1) \cdot a + a^2$$

Take the expectation (AKA average) denoted

$$\langle x(t)^2 \rangle = \langle x(t-1)^2 \rangle + \langle x(t-1) \cdot a \rangle + \langle a^2 \rangle$$

We can distribute the angle brackets because the average of the sum is the sum of the averages (this is called linearity of expectation).

The steps $\pm a$ are independent of $x(t-1)$ and $\langle a \rangle = 0$ because our walker is equally likely to walk forwards or backwards. So

$$\langle x(t-1) \cdot a \rangle = \langle x(t-1) \rangle \cdot \langle a \rangle = 0$$

Finally, since a is a constant, $\langle a^2 \rangle = a^2$ and

$$\langle x(t)^2 \rangle = \langle x(t-1)^2 \rangle + a^2.$$

Since $\langle x(0)^2 \rangle = 0$ and we accumulate a unit of a^2 at each step, we see that $\langle x(t)^2 \rangle = a^2 \cdot t$.

9.2 A binomial distribution approach to $l^2 \propto t$ in 1D

Suppose we move right when we toss heads with probability p (coin might be unfair). What is the probability of n_R rightward steps in N total? This value is expressed by the binomial distribution

$$p(n_r|N) = \binom{N}{n_r} p^{n_r} (1-p)^{N-n_r}$$

Here p^{n_r} is the probability of getting at least n_r heads, and $(1-p)^{N-n_r}$ is the probability of rolling $N - n_r$ tails. This product is the probability of a string of coin tosses with exactly n_r heads. However this string can come in many different orders (multiplicities). The binomial coefficient

$$\binom{N}{n_r} = \frac{N!}{n_R!\,(N-n_R)!}$$

gives the number of ways (orderings) of choosing n_r heads from N tosses.

A string of tosses has a displacement

$$x = a(n_r - n_l) = a(2n_r - N).$$

This string has probability $p^{n_r}(1-p)^{N-n_r}$, but there are $\binom{N}{n_r}$ strings with the same number of total steps N and rightward steps n_r. All of these have the same x.

You can try using the binomial distribution to

1. Calculate the average displacement $\langle x \rangle$ for N tosses.
2. Calculate the average squared displacement $\langle x^2 \rangle$

10 Scaling of RMSD with t

We saw that random motion implies a linear relationship of $\langle x(t)^2 \rangle \propto Dt$ via dimensional analysis. In 3D, $\langle x(t)^2 \rangle = 6Dt$. If we plot $\langle x^2 \rangle$ against t, as in Fig. 6, then the slope is $6D$.

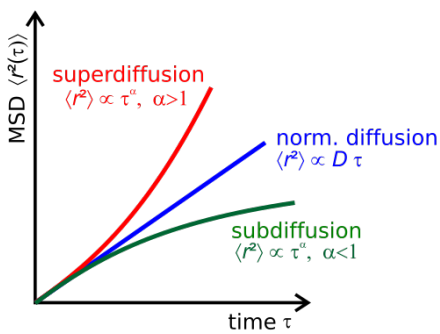

Figure 6: Diffusion implies a linear relationship of x^2 with t. From Wikipedia on Anomalous Diffusion.

This gives us a way of measuring D by observing molecules over time (as Perrin did). But it also gives us something more deep than that – **a way of diagnosing whether the underlying motion is diffusive in nature**. If we plot $\langle x^2 \rangle$ against t and we don't get a line, then the motion is not diffusive!

10.1 Measuring D by scaling

This experiment that was done in the lab of Christine Jacobs-Wagner (Parry et al. Cell 2014). They made fluorescent particles (μNS) in bacteria and tracked them over time. They are large and dilute, so they can be tracked individually, much like Perrin's colloidal particles.

How could we determine D from such traces? We would measure the displacement x as a function of t. Each individual trajectory is random, but the **average** behavior is $\langle x^2 \rangle \propto Dt$ as in Figure 6.

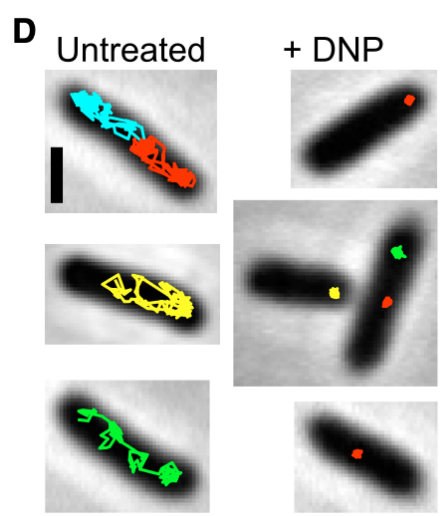

Figure 7: Traces of μNS particles in treated and untreated cells. Parry et al., Cell 2014.

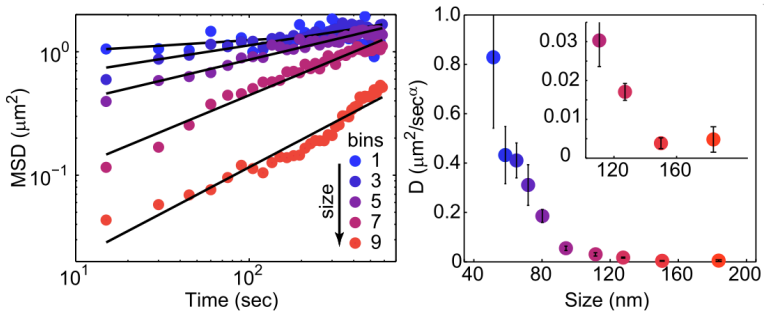

Figure 8: MSD traces for particles of distinct size measured in *E. coli* lysate. From Fig. S6 of Parry et al. Cell 2014

When the data fit a line well, then we take the slope, which equals $4D$ in two dimensions (the Parry et al. case) and $6D$ in 3D. **What if you don't see a line?** What can you conclude?

10.2 Estimate D from data

Kubitscheck et al. 2000 tracked GFP particles in a viscous solution (80% glycerol). From Figure 9, we estimate

$$4D \approx \frac{1.0\mu\mathrm{m}^2}{0.1\mathrm{s}} \approx 10\mu\mathrm{m}^2/\mathrm{sec}$$

Giving $D \approx f\mu\mathrm{m}^2/\mathrm{sec}$. The measurement was conducted at a viscosity of ≈ 50 cPoise while cytoplasm has a viscosity around $1-2$ cPoise. So we expect GFP to have a diffusion constant $D \approx 10 - 100\mu\mathrm{m}^2/\mathrm{sec}$.

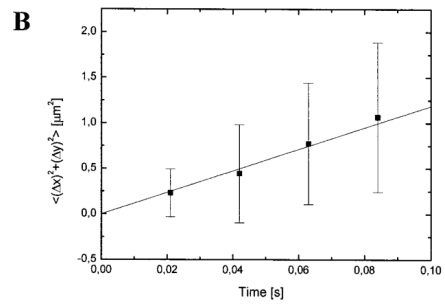

Figure 9: Single particle tracking of GFP in a viscous solution, Kubitscheck et al. Biophys J. 2000

10.3 Other modes of measurement

How might you measure x^2 vs t to infer D? Ideas:

1. Single particle tracking.
2. Follow a spreading bolus injected at a point.
3. Removing signal (e.g., by bleaching) and monitoring recovery (FRAP).
4. Add signal locally, e.g., by photoactivation (Elowitz JBac 1999)

There are also other approaches involving correlation spectroscopy. It's worth learning about these methods because they rely on different aspects of diffusion to make the measurement – have different deficits and benefits.

11 Estimating the diffusion time t_{diff}

Table 2 gives typical values of D for some interesting molecules/complexes. **Take a minute** to estimate the diffusion time for GFP across a bacterium

$$t_{diff} = \frac{l^2}{6D} \approx \frac{1\mu\mathrm{m}^2}{6 \times 30\mu\mathrm{m}^2\,\mathrm{s}^{-1}} \approx f \text{ ms.}$$

Repeat this calculation for glucose (0.f-1 ms) and for GFP in an axon of with length ≈ 1 cm $= 10^4$ micron. For the axon

$$t_{diff} = \frac{l^2}{6D} \approx \frac{(10^4)^2\mu\mathrm{m}^2}{6 \times 30\mu\mathrm{m}^2\,\mathrm{s}^{-1}} \approx f \times 10^5 \text{ s.}$$

Molecule	D ($\mu\mathrm{m}^2\,\mathrm{s}^{-1}$)
H_2O in water	2000
CO_2 in water	2000
Glucose in water	600
GFP in water	100
GFP in cytoplasm	30

Table 2: D for select molecules (from the book).

A day has $24 \times 3600 = 84,000 \approx 10^5$ seconds, so it would take GFP a few days to cross an axon by diffusion.

Rule of Thumb: A day in seconds

1 day $\approx 10^5$ seconds

12 Perfect absorber

Consider a bacterium in a dilute medium with ≈ 10 mM glucose, like in the lab. We started by saying it needs $\approx 10^6$ sugars/s to support a 1 hr generation. **Will it actually encounter this number of sugars by diffusion?**

How should we approach this problem? Treat the cell as a sphere of radius a that absorbs all arriving sugar molecules (Fig. 10). It the surface receives a sugar flux $J(a) \cdot SA = 4\pi a^2 J(a)$. Mass conservation says $4\pi R^2 J(R)$ is constant for all R, so

$$J(R) \propto \frac{1}{R^2}$$

Fick's first law tells us

$$J(R) = -D\frac{dc}{dR} \implies \frac{dc}{dR} = \frac{A}{R^2}$$

This gives

$$c(r) = \frac{A}{R} + B; \qquad c(a) = 0; \qquad c(\infty) = c_\infty = 10\text{mM}$$

Here we absorbed a minus sign into A and we wrote our boundary conditions knowing the bulk sugar concentration (10 mM) and that the perfect absorber consumes everything at its surface. Plugging the boundary conditions in gives

$$c(\infty) = c_\infty = B; \qquad c(a) = 0 = \frac{A}{a} + c_\infty \implies A = -c_\infty \cdot a$$

And

$$c(R) = c_\infty(1 - \frac{a}{R})$$

Note: when teaching, we **can posit this form** by inspecting the boundary conditions.

Now we can calculate the flux at the cell surface

$$J(R=a) = -D\frac{dc(R)}{dR}\bigg|_{R=a} = -D\frac{a}{R^2}c_\infty\bigg|_{R=a} = -D\frac{c_\infty}{a}$$

OK, now we can estimate.

$$\begin{aligned} J(a) \cdot SA &= -D\frac{c_\infty}{a} \cdot 4\pi a^2 \\ &= -Dc_\infty \cdot 4\pi a \end{aligned}$$

We chose $a = 1\mu\text{m}$ so $4\pi a \approx 10\mu\text{m}$

$$\approx 600\mu\text{m}^2\,\text{s}^{-1} \times 10\mu\text{m} \times 10^{-2}\frac{\text{mol}}{\text{L}} \times \frac{1\text{L}}{1\text{dm}^3} \times \left(\frac{1\text{dm}}{10^5\mu\text{m}}\right)^3 \times \frac{10^{24}\text{molecules}}{\text{mol}}$$

$$\approx 6 \times 10^{10}\frac{\text{glucose}}{\text{s}}$$

the diffusion-limited on-rate

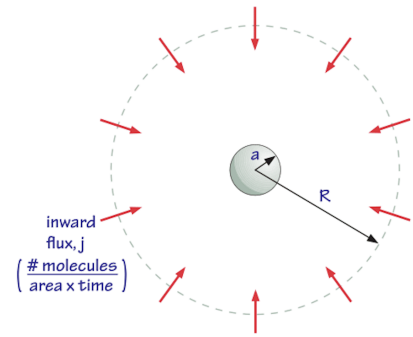

Figure 10: setup for the perfect absorber from the book.

Note: this approach was first described in a classic 1997 paper by Berg and Purcell. Since we rounded several values, you may see estimates that are a bit lower ($\sim 10^9$ glucose/s) in books/papers.

13 Teaser: diffusion limited on-rate

The rate of a reaction is often treated as 'first-order' and described with rate constant k_{on} that multiplies the substrate concentration c.

$$v = E \cdot k_{on} \cdot c.$$

Here v has units of [M/s], k_{on} has units of [/M/s], and the enzyme concentrations E and c have molar units. The per-enzyme rate

$$v/E = k_{on} \cdot c$$

reflects the rate at which substrates arrive at the enzyme. If an enzyme performs the catalytic steps very quickly, then k_{on} will be the rate at which substrates arrive at the enzyme by diffusion.

What is the fastest enzyme known?

1. $k_{on} \approx 10^3$ /M/s
2. $k_{on} \approx 10^5$ /M/s
3. $k_{on} \approx 10^7$ /M/s
4. $k_{on} \approx 10^9$ /M/s
5. $k_{on} \approx 10^{11}$ /M/s

Answer $10^9 - 10^{10}$ /M/s for triose-phosphoate isomerate (TIM) and superoxide dismutase (SOD, Figure 11)

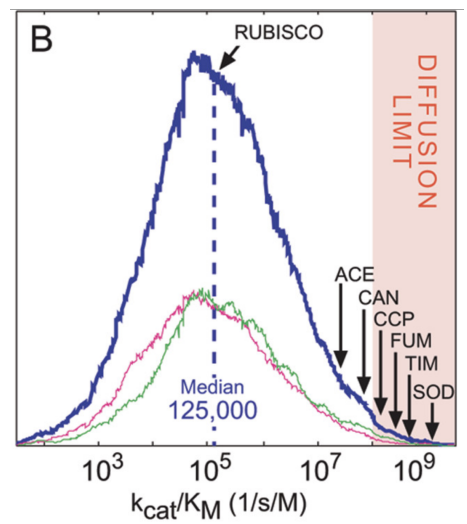

Figure 11: Empirical distribution of $k_{on} \approx k_{cat}/K_M$ from Bar-Even et al. Biochem 2011.

13.1 Lessons for thinking about metabolism

Thoughts? The diffusion limit on enzymatic catalysis is a kind of speed limit on metabolism, signaling, and other cellular processes catalyzed by enzymes.[2] If cells want to go faster, they have to pull tricks. What are some plausible ways of going faster?

1. Express more enzyme to get more total rate – this too has limits.
2. For an enzyme cascade $S \underbrace{\longrightarrow}_{E_1} I \underbrace{\longrightarrow}_{E_2} P$ co-localize E_1 and E_2.
3. Similar: active transport of substrate to enzyme (or vice versa)
4. The substrate could evolve to be smaller to diffuse faster.
5. Animals+plants have circulatory systems that bring substrates to cells + enzymes.

13.2 Relationship with the Michaelis-Menten rate equation

You might be familiar with the Michaelis-Menten rate equation for enzymes:

$$v = E \cdot \frac{k_{cat} \cdot c}{K_M + c}.$$

In the dilute limit $c \ll K_M$ we recover the 'first-order' kinetic equation above with $k = k_{cat}/K_M$ and $v = \frac{k_{cat}}{K_M} \cdot c$. This is the limit we are interested in. In the high concentration limit ($c \gg K_M$) the substrate is so concentrated that the total rate $v = k_{cat}$ is limited by the enzymes ma 'diffusion-limited' Indeed, this is the limit we are interested in. In

[2] Protein kinases and polymerases are also enzymes.

13.3 Estimating the diffusion-limited k_{on}

Let's consider an enzyme using the "perfect absorber" approach from above. Consider an enzyme, which is a few hundred times smaller than a cell. We want to calculate the on-rate, which multiplies the bulk substrate concentration

$$k_{on}c_\infty = Dc_\infty \cdot 4\pi a.$$

We'll use $D = 100\ \mu\mathrm{m}^2\,\mathrm{s}^{-1}$ as a typical small molecule diffusion constant and $a = 2$ nm $= 2 \times 10^{-3}$ $\mu\mathrm{m}$ as a typical radius for a monomeric protein. This gives

$$\begin{aligned} k_{on} &\approx 4\pi \times 10^2 \mu\mathrm{m}^2\,\mathrm{s}^{-1} \times 2 \times 10^{-3}\mu\mathrm{m} \times \frac{1\mathrm{L}}{10^{15}\mu\mathrm{m}^3} \times 6 \times 10^{23}\frac{\text{molecules}}{\mathrm{mol}} \\ &\approx 150 \times 10^7 \approx 10^9\ /\text{M/s} \end{aligned}$$

Some enzymes might be larger. **What would this do to our estimate?** Similarly, what would change if D was larger or smaller, as it is for glucose. Since these factors vary from enzyme to enzyme, we often see a the diffusion limited on-rate quoted as $10^9 - 10^{10}$ /M/s (see Bar-Even et al Biochem. 2015). The book has a nice vignette on this topic and the original derivation is due to Alberty and Hammes (J. Phys. Chem 1958).

14 Estimating D for a spherical particle

We want to find the diffusion coefficient D for a spherical particle with radius r. We'll try dimensional analysis to learn how D scales with size r. D has dimensions:

$$[D] = L^2T^{-1}$$

Available physical quantities: For a sphere of radius r diffusing in a fluid of viscosity η at temperature T:

Quantity	Symbol	Dimensions
Particle radius	r	L
Fluid viscosity	η	$ML^{-1}T^{-1}$
Thermal energy	k_BT	ML^2T^{-2}

We are balancing the thermal energy of our spherical particle it's "jiggliness" – with the fluid viscosity – the degree to which a fluid resists motion.

The thermal energy scale (k_BT) has dimensions of energy, e.g., units of joules. 1 J = 1 N·m = 1 kg m^2/s^2. Near 20 C, water has a viscosity of ≈ 1 cPoise = 1 mPa · s = 10^{-3} kg m^{-1} s^{-1}.

The viscosity dimensions follow from its definition

$$\tau = \frac{F}{A} = \eta\frac{\partial v}{\partial z}$$

That is, $\eta =$ stress/strain rate $= (F/A)/(\partial v/\partial z)$. F is a force with dimensions $M \cdot L \cdot T^{-2}$, A an area with dimensions L^2 and the strain rate $(\partial v/\partial z)$ has dimensions T^{-1}. η therefore has dimensions of $ML^{-1}T^{-1}$.

14.1 Dimensional Analysis for η

We seek $D \sim (k_B T)^\alpha \cdot \eta^\beta \cdot r^\gamma$. Writing dimensions:

$$L^2 T^{-1} = (M L^2 T^{-2})^\alpha \cdot (M L^{-1} T^{-1})^\beta \cdot L^\gamma$$

Matching powers of M, L, T:

$$\begin{aligned} M: &\quad 0 = \alpha + \beta \\ L: &\quad 2 = 2\alpha - \beta + \gamma \\ T: &\quad -1 = -2\alpha - \beta \end{aligned}$$

From the M equation: $\beta = -\alpha$. Substituting into T: $-1 = -2\alpha + \alpha$, so $\alpha = 1$, $\beta = -1$. Then from L: $\gamma = 2 - 2 + (-1) = -1$, giving:

$$D \sim \frac{k_B T}{\eta r} \tag{3}$$

14.2 Full Stokes-Einstein expression for D

The full Stokes-Einstein result is:

$$D = \frac{k_B T}{6\pi\eta r}$$

The 6π prefactor comes from solving the Stokes drag problem explicitly — the drag force on a sphere is $F = 6\pi\eta r v$, giving a friction coefficient $\gamma = 6\pi\eta r$, which enters via the Einstein relation $D = k_B T/\gamma$. Dimensional analysis fixes the functional form but not the numerical prefactor.

Can we estimate the values in Table 2?

14.3 Estimation of D_{GFP} with Stokes-Einstein

GFP is has a hydrodynamic radius of a f nm $= f \times 10^{-9}$ m ([BNID 104396]).

- Boltzmann constant $k_B = 1.38 \times 10^{-23}$ J K^{-1}
- Absolute temperature $T \approx 300$ K $(25 - 30$ C$)$
- Therman energy $k_B T \approx f \times 10^{-21}$ J $= f \times 10^{-21}$ kg m^2 s^{-2}
- Water-like viscosity $\eta \approx 1$ cPoise $= 1$ mPa $\cdot$ s $= 10^{-3}$ kg m^{-1} s^{-1}.
- GFP hydrodynamic radius $r = f \times 10^{-9}$ m

Plugging these values Eq. 14.2

$$D = f \times 10^{-21} \mathrm{kg\,m^2\,s^{-2}} \times \frac{1}{6\pi \times f \times 10^{-9} \mathrm{m}} \times \frac{1}{10^{-3} \mathrm{kg\,m^{-1}\,s^{-1}}} \approx f \times 10^{-11} \mathrm{m^2\,s^{-1}}$$

Converting this to $\mu\mathrm{m}^2\,\mathrm{s}^{-1}$ gives

$$f \times 10^{-11} \mathrm{m^2\,s^{-1}} \times \left(\frac{10^6 \mu\mathrm{m}}{1\mathrm{m}}\right)^2 = f \times 10\,\mu\mathrm{m^2\,s^{-1}}$$

Pretty good.

14.4 Scaling by size to estimate D_{GFP}

Another way of getting here is to remember the diffusion coefficient of water in water, which is about $2000\ \mu\text{m}^2\,\text{s}^{-1}$ (Table 2). D for similarly-sized molecules like CO_2 O_2 are about the same.

Consider that $D \propto r^{-1}$ so that

$$\frac{D_{\text{GFP}}}{D_{\text{water}}} = \frac{r_{\text{water}}}{r_{\text{GFP}}} \approx \frac{0.1\text{nm}}{f\text{nm}} \approx f \times 10^{-2}$$

This gives

$$D_{\text{GFP}} \approx f \times 10^{-2} \times D_{\text{water}} \approx 50 - 100\mu\text{m}^2\,\text{s}^{-1}$$

Not bad either.

14.5 Ballistic motion

When you walk, bike or drive, your motion is not random. It's directed, or ballistic, like a bullet or rocket. These objects a velocity v and the distance traveled *scales linearly with time*:

$$d = v \cdot t$$

if we plot d^2 against t it will not form a line. **What would the plot look like? Try to sketch it.**

Let's start simple: $d(t = 0) = 0$. For $t > 0$, $d^2 \propto t^2$. So we are plotting $d^2 = y(t) = v^2 \cdot t^2$. This will give a concave-upwards plot like the red curve in Figure 6.

What are some biological examples of ballistic motion?

1. Motor proteins on actin filaments / microtubules
2. Circulatory systems – blood, xylem/phloem
3. Bikes, cars, skateboards, airplanes.

Actually, at the macro-scale it's hard to think of examples of diffusive motion. The most accessible examples invovle adding dyes (e.g., food coloring) to liquids.

But if we consider a neuronal axon, which might be $100\,\mu\text{m}$ we start to see a challenge

$$t_{diff} = \frac{(10^2)^2\mu\text{m}^2}{6 \cdot f \times 10\mu\text{m}^2\,\text{s}^{-1}} \approx 10^2\text{second}.$$

For longer distances or slower diffusing (larger) molecules, diffusion is just too slow to support life.

If we're thinking about *E. coli* or yeast cells, the relevant length scales are $1 - 10\mu\text{m}$. Neuron

Let's compare a small molecule like glucose ($D \approx f \times 10^2\mu\text{m}^2\,\text{s}^{-1}$) to a globular protein like GFP ($D = f \times 10\mu\text{m}^2\,\text{s}^{-1}$ in cytoplasm).

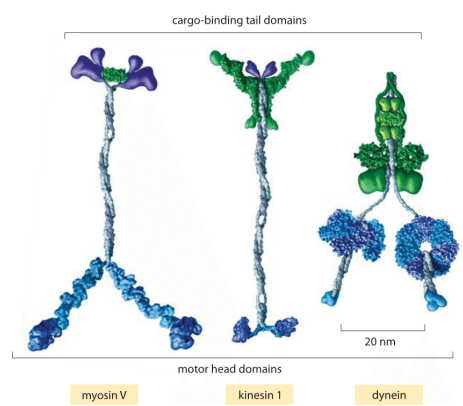

Figure 12: Structures of molecular motors

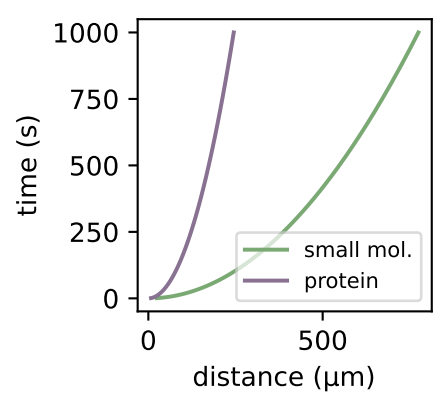

Figure 13: Scaling of distance traveled with time for diffusive motion.

15 Advective transport and the Peclet number

Animal and plant bodies have circulatory systems that move fluids through the body. The moving fluid carries circulating nutrients (glucose, lactate, LDL) and cells (e.g., red blood cells, neutrophils) around the body. This "advective" mode of transport requires energy input (e.g., from the heart).

The aorta is the bodies largest artery, being about 30 cm long (Figure 15). **How long would it take to for a protein cross 30 cm** by diffusion? We know that this will be very slow because we already made a calculation at the 1 cm scale.

$$t_{diff} = \frac{l^2}{6 \cdot D} = \frac{(f \times 10 \text{ cm})^2}{6 \cdot f \times 10\mu\text{m}^2\,\text{s}^{-1}} \approx \frac{(f \times 10^5\mu\text{m})^2}{6 \cdot f \times 10\mu\text{m}^2\,\text{s}^{-1}} \approx 10^9 \text{ s.}$$

A nice rule of thumb is that there are $365 * 24 * 3600 = 3.15 * 10^7 \approx \pi \times 10^7$ s/yr, meaning that protein diffusion across the aorta would take ≈ 30 years.

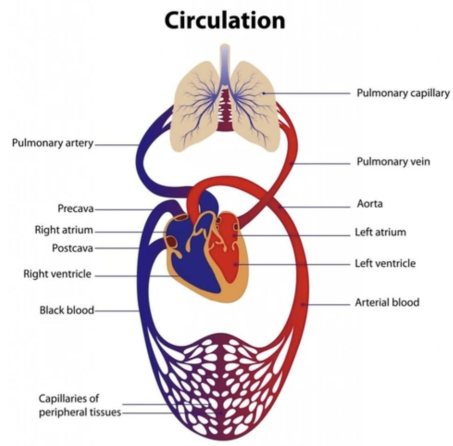

Figure 14: The human circulatory system.

15.1 Do I run faster than my blood flows?

How fast does blood flow? It will of course depend on the vessel or artery we consider, so lets consider the aorta, which is the largest artery in humans.

One approach to estimation: adult humans have a few liters of blood. The blood needs to circulate fast enough to supply a whole body with O_2.

Rule of Thumb: Blood Volume

$\approx 60 - 70\,\text{mL}\,\text{kg}^{-1}$ for adult humans

An adult human consumes $\approx f \times 10^2$ ml O_2 per minute.[3] Using the numbers in Table 3, we estimate that blood carries

$$\frac{1.3 \text{ mL O2}}{\text{g hemoglobin}} \times \frac{15\text{g hemoglobin}}{100 \text{ mL blood}} \approx \frac{20 \text{ mL O2}}{100 \text{ mL blood}} = \frac{200 \text{ mL O2}}{\text{L blood}}$$

Quantity	Value	Units
O_2 consumption (adult at rest)	$\approx f \times 10^2$	ml O_2 / min
O_2 carried per gram hemoglobin	≈ 1.3	ml O_2 / g Hgb
Blood hemoglobin concentration	15	g Hgb / 100 ml blood

Table 3: Numerical values from the passage on oxygen transport in blood from Medical Physiology by Hall & Hall.

We can now estimate the volume of blood needed to supply and adult with 1 minutes worth of O_2.[4]

$$2 \times f \times 10^2\,\text{mL}\,\text{min}^{-1}\ \text{O}_2 \times \frac{1\text{L blood}}{2 \times 10^2\,\text{mL O}_2} \approx \frac{f\,\text{L blood}}{\text{min}}$$

The leading factor of 2 is needed because, for every L of blood carrying O_2, there is another that is deoxygenated. So you have a few L of blood and they

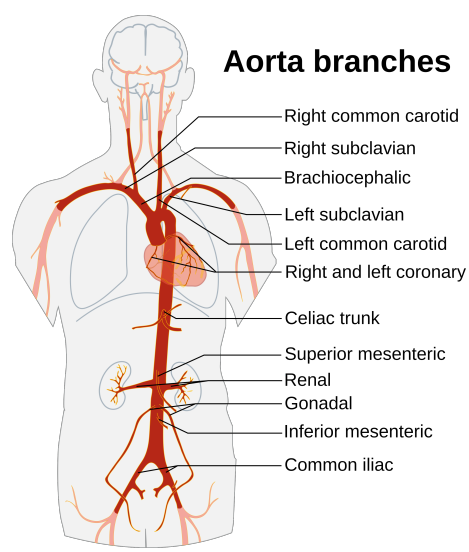

Figure 15: Branches of the human aorta (Wikipedia).

[3] A mL of O_2 is a number of moles of O_2 that would fill 1 mL at STP. Also, we could try to estimate this by approximating our breaths per minute, breath volume, and knowing the O_2 content of air.

[4] Add blood oxygen carrying capacity to Bionumbers?

circulate through your body every minute or so (this is the "blood circulation time").

15.1.1 Speed of flow in the aorta

All the blood in your body passes through your aorta. At what speed does it flow? A human aorta has a radius $r \approx 1$ cm, a and a cross-sectional area $A = \pi r^2 \approx f$ cm^2.

We have a cardiac output of $Q = f$ L/min $= f \times 10^3$ cm^3/min and a cross-sectional area of $A = f$ cm^2. Considering the units, we see that the flow speed

$$v = \frac{Q}{A} = \frac{f \times 10^3 \text{ cm}^3/\text{min}}{f \text{ cm}^2} \times \frac{1 \text{ min}}{60 \text{ s}} \approx f \times 10 \text{ cm/s}$$

To see this another way, consider a reference length of aorta, say $l_{ref} = 1cm$. The reference volume $V_{ref} = A \cdot l$. The ratio Q/V gives the volume the aorta must carry per minute, so the velocity

$$v = l_{ref} \cdot \frac{Q}{V_{ref}} = \frac{Q}{A}$$

Is this flow fast compared to my (very slow) running speed of 5 mph ≈ 2 m/s? Well, **no**: $f \times 10$ cm/s $\approx f \times 10^{-1}$ m/s.

16 Fick's second law

Fick's first law relates the the diffusive flow of molecules (J) to the concentration profile in space ($\frac{dc}{dx}$). Fick's second law is different and more challenging. It relates the change in concentration over time at a fixed location ($\frac{dc}{dt}\big|_x$) to the second derivative of the concentration in space $\frac{d^2c}{dx^2}$. I find this statement pretty confusing, but it's also powerful and worth your time. Consider a box at x with area $A = l^2$ and width δx as in Figure 16. The number of molecules in the box is $N_x = c(x,t) \cdot l^2 \cdot \delta x$. N_x increases if molecules enter the box, and decreases if they leave. So, over a time interval Δt,

$$\begin{aligned} N_x(t+\Delta t) - N_x(t) &= \underbrace{J_x \cdot A \cdot \Delta t}_{\text{entering}} - \underbrace{J_{(x+\delta x)} \cdot A \cdot \Delta t}_{\text{leaving}} \\ &= (J_x - J_{(x+\delta x)}) \cdot A \cdot \Delta t \end{aligned}$$

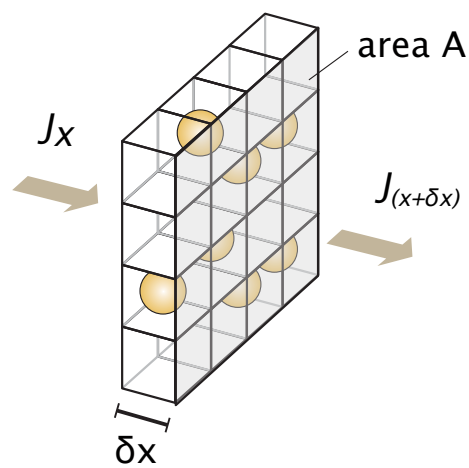

Figure 16: A volume with width δx and area A gains and loses molecules via diffusive influx from the left and outflux to the right.

Notice that the LHS reads $N_x(t+\Delta t) - N_x(t)$ because N_x is increasing when $J_x - J_{x+\delta x} > 0$ while N_x is decreasing when $J_x - J_{x+\delta x} < 0$. This is simply due to the orientation of flow assumed in Fig. 16.

Rearranging and dividing both sides by δx then we get

$$\begin{aligned} \frac{N_x(t+\Delta t) - N_x(t)}{A \cdot \delta x \cdot \Delta t} &= \frac{J_x - J_{(x+\delta x)}}{\delta x}, \\ -\frac{dc(x,t)}{dt} &= \frac{dJ_x}{dx} \end{aligned}$$

in the limit of small δx and Δt. Since Fick's first law gives flux $J_x = -D\frac{dc(x,t)}{dx}$ we can now derive Fick's second law:

$$\frac{dc(x,t)}{dt} = D\frac{d^2c(x,t)}{dx^2}.$$

We see that Fick's second law relates the time derivative of concentration $c(x,t)$ to the change in flux, which is proportional to the second derivative of $c(x,t)$ in diffusion (in the absence of reactions).

17 Lecture 4 Handout

17.1 Brown on Molecular Jiggling

> The grains of pollen ... were filled with particles or granules of unusually large size, varying from nearly $\frac{1}{4000}$th to about $\frac{1}{3000}$th of an inch in length ... I observed many of them very evidently in motion.[5]
> — Robert Brown, 1827

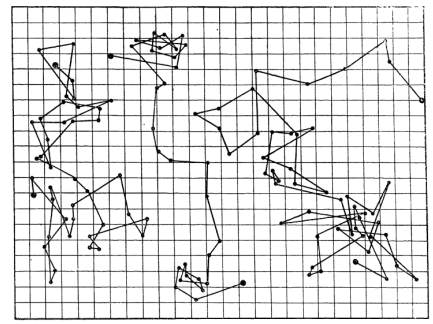

Figure 17: Tracings of the motions of three colloidal particles of radius 0.53 microns as seen under the microscope (Perrin 1909). Each point representing that particle's successive position every 30 seconds; the points are then joined by straight line segments. Caption text reproduced from Wikipedia on Brownian motion

17.2 On Null Models

> A mathematical model is neither an hypothesis nor a theory. Unlike the scientific hypothesis, a model is not verifiable directly by experiment. For all models are both true and false. Almost any plausible propose relation among aspects of nature is likely to be true in the sense that it occurs (although rarely and slightly). Yet all models leave out a lot and are in that sense false, incomplete, inadequate. The validation of a model is not that it is "true" but that it generates good testable hypotheses relevant to important problems.
> — Richard Levins, "Model Building in Pop. Bio." 1966

17.3 Sanjoy M. Comments on Units and Dimensions

Sanjoy Mahajan offers the following quote in Ch. 1 of "Streetfighting Mathematics"

> In Nigeria, a relatively economically strong country, the GDP [gross domestic product] is $99 billion. The net worth of Exxon is $119 billion. "When multinationals have a net worth higher than the GDP of the country in which they operate, what kind of power relationship are we talking about?" asks Laura Morosini.
> – Anne Marchand. Impunity for Multinationals. September 2002.

[5] So how big are pollen grains? $\frac{1 \text{ inch}}{f \times 10^3} \approx \frac{f \text{ cm}}{f \times 10^3} = 10^{-5}$ m or about 10 microns in diameter

18 Diffusion Figures

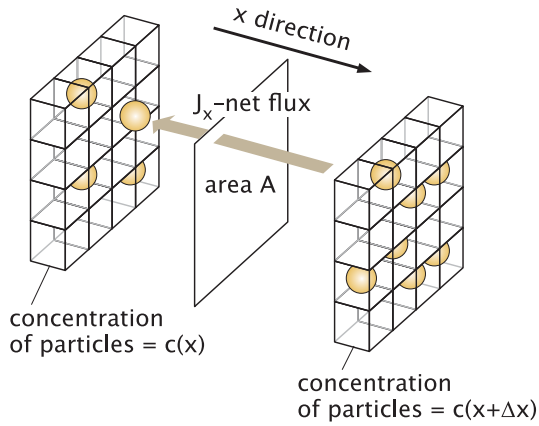

Figure 18: Random motion of a molecule between adjacent planes.

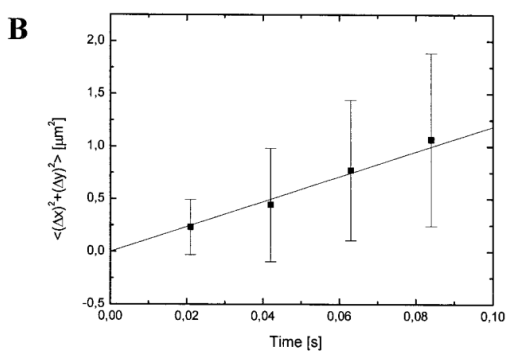

Figure 19: Single particle tracking of GFP in a viscous solution, Kubitscheck et al. Biophys J. 2000

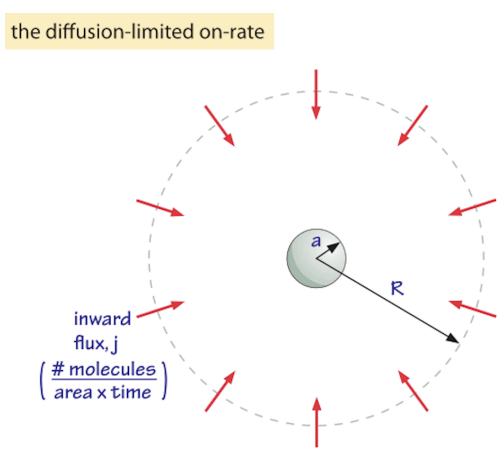

Figure 20: Setup for the perfect absorber calculation from the book.